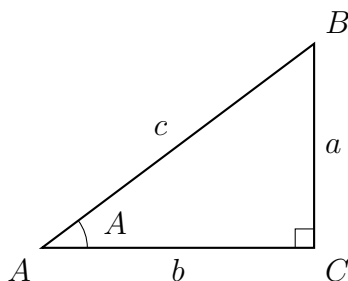


Finding the Error in Right-Triangle Trigonometry

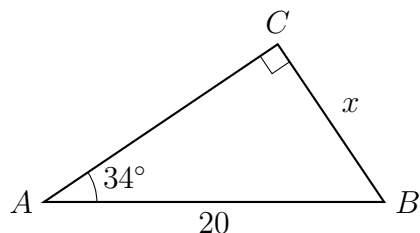
Directions

- Each problem shows a calculation that produced a wrong answer. The arithmetic is done correctly every time — the fault is the ratio chosen, the inverse function chosen, or the calculator's angle mode.
- Redo the calculation correctly and write the value, with its unit, on the answer line. Round lengths to two decimal places and angles to two decimal places.
- Then name the error in a sentence, and where you can, say what about the wrong answer should have given it away.
- Every diagram is drawn to scale from that problem's givens.



How every triangle here is labelled. No values shown: use the numbers given in each problem.

1. In the right triangle below the hypotenuse is 20 cm and $m\angle A = 34^\circ$. To find the side opposite $\angle A$, a student computed $20 \cos 34^\circ = 16.58$ cm.

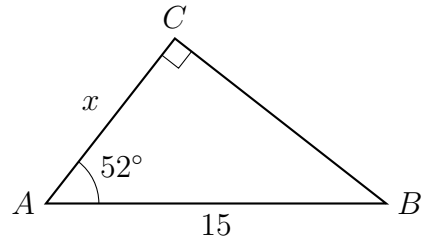


- (a) Find the correct length of the side opposite $\angle A$, in centimetres.

Answer: _____ cm

- (b) Name the error in the student's choice of ratio.

2. In the right triangle below the hypotenuse is 15 in and $m\angle A = 52^\circ$. To find the side adjacent to $\angle A$, a student entered $15 \cos 52$ and the calculator returned -2.44 in.

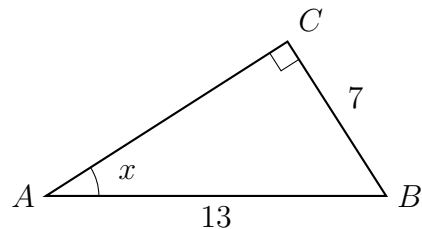


- (a) Find the correct length of the side adjacent to $\angle A$, in inches.

Answer: _____ in

- (b) Name the error, and say what about the printed answer makes it impossible.

3. In the right triangle below the side opposite $\angle A$ is 7 cm and the hypotenuse is 13 cm. A student wrote $m\angle A = \tan^{-1}(7/13) = 28.30^\circ$.

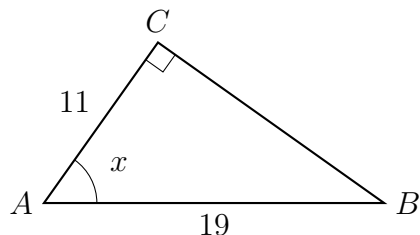


- (a) Find the correct value of $m\angle A$, in degrees.

Answer: _____ degrees

- (b) Name the error in the student's choice of inverse function.

4. In the right triangle below the side adjacent to $\angle A$ is 11 m and the hypotenuse is 19 m. A student divided 11 by 19 and reported $m\angle A = 0.58^\circ$.

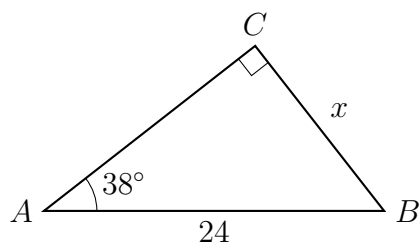


- (a) Find the correct value of $m\angle A$, in degrees.

Answer: _____ degrees

- (b) Name the error, and say what about the printed answer makes it impossible.

5. A 24 m cable runs from the ground to the top of a pole, making an angle of 38° with the ground. To find how high up the pole the cable reaches, a student entered $24 \sin 38$ and got 7.11 m.

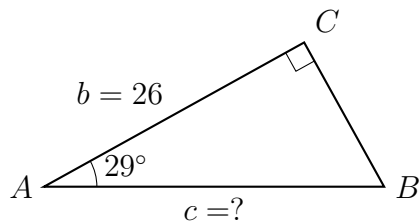


- (a) Find the correct height, in metres.

Answer: _____ m

- (b) Name the error, and give a rough estimate that would have exposed it.

6. In the right triangle below $m\angle A = 29^\circ$ and the side adjacent to $\angle A$ is 26 ft. To find the hypotenuse, a student computed $26 \tan 29^\circ = 14.41$ ft.



(a) Find the correct length of the hypotenuse, in feet.

Answer: _____ ft

(b) Name the error, and say what about the printed answer makes it impossible.